

API Documentation & Submission Notes

Spring Boot · H2 (in-memory) · Swagger / OpenAPI

ENDPOINTS

Method	Endpoint	Description
POST	/api/tickets	Creates a support ticket and queues it for AI analysis.
GET	/api/tickets/{id}	Retrieves a ticket by ID, including the AI analysis once processing completes.

REQUEST / RESPONSE EXAMPLES

POST /api/tickets

REQUEST BODY

```
{
  "customerId": "CUST-301",
  "subject": "Password reset link not working",
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth"
}
```

RESPONSE — 201 CREATED

```
{
  "id": 1,
  "customerId": "CUST-301",
  "subject": "Password reset link not working",
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth",
  "status": "PENDING",
  "failureReason": null,
  "analysis": null,
  "createdAt": "2026-09-27T11:19:08.435Z",
  "updatedAt": "2026-09-27T11:19:08.435Z"
}
```

Analysis runs asynchronously — the ticket is created and returned immediately with status: PENDING, before the AI analysis has run.

GET /api/tickets/{id}

RESPONSE — 200 OK (after analysis completes)

```
{
  "id": 1,
  "customerId": "CUST-301",
  "subject": "Password reset link not working",
```

```
{
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth",
  "status": "COMPLETED",
  "failureReason": null,
  "analysis": {
    "category": "ACCOUNT_ACCESS",
    "summary": "Customer reported an issue with subject \"Password reset link not working\" related to Auth.",
    "suggestedResponse": "Thank you for reaching out. We've received your report and our account security team is looking into it. We'll follow up as soon as we have an update.",
    "recommendedTeam": "ACCOUNT_SECURITY",
    "confidence": 0.85
  },
  "createdAt": "2026-09-27T11:19:08.435Z",
  "updatedAt": "2026-09-27T11:19:08.829Z"
}
```

SWAGGER DOCUMENTATION: POST:

POST /api/tickets

Parameters

No parameters

Request body `model`

application/json

```
{
  "customerId": "C001-001",
  "subject": "Password reset link not working",
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth",
  "status": "COMPLETED",
  "failureReason": null,
  "analysis": {
    "category": "ACCOUNT_ACCESS",
    "summary": "Customer reported an issue with subject \"Password reset link not working\" related to Auth.",
    "suggestedResponse": "Thank you for reaching out. We've received your report and our account security team is looking into it. We'll follow up as soon as we have an update.",
    "recommendedTeam": "ACCOUNT_SECURITY",
    "confidence": 0.85
  },
  "createdAt": "2026-09-27T11:19:08.435Z",
  "updatedAt": "2026-09-27T11:19:08.829Z"
}
```

Execute Clear

Responses

curl

```
curl -X POST \
  http://localhost:8080/api/tickets/ \
  -H 'accept: */*' \
  -H 'content-type: application/json' \
  -d '{
    "customerId": "C001-001",
    "subject": "Password reset link not working",
    "description": "I clicked the password reset link in my email but it says the link has expired",
    "priority": "MEDIUM",
    "product": "Auth",
    "status": "COMPLETED",
    "failureReason": null,
    "analysis": {
      "category": "ACCOUNT_ACCESS",
      "summary": "Customer reported an issue with subject \"Password reset link not working\" related to Auth.",
      "suggestedResponse": "Thank you for reaching out. We've received your report and our account security team is looking into it. We'll follow up as soon as we have an update.",
      "recommendedTeam": "ACCOUNT_SECURITY",
      "confidence": 0.85
    },
    "createdAt": "2026-09-27T11:19:08.435Z",
    "updatedAt": "2026-09-27T11:19:08.829Z"
  }'
```

Response URL

http://localhost:8080/api/tickets

Server response

Code Details

201

Response body

```
{
  "id": 1,
  "customerId": "C001-001",
  "subject": "Password reset link not working",
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth",
  "status": "COMPLETED",
  "failureReason": null,
  "analysis": {
    "category": "ACCOUNT_ACCESS",
    "summary": "Customer reported an issue with subject \"Password reset link not working\" related to Auth.",
    "suggestedResponse": "Thank you for reaching out. We've received your report and our account security team is looking into it. We'll follow up as soon as we have an update.",
    "recommendedTeam": "ACCOUNT_SECURITY",
    "confidence": 0.85
  },
  "createdAt": "2026-09-27T11:19:08.435Z",
  "updatedAt": "2026-09-27T11:19:08.829Z"
}
```

Response headers

```
connection: keep-alive
content-type: application/json
date: Sat, 27 Sep 2025 11:20:08 GMT
keep-alive: timeout=60
transfer-encoding: chunked
```

GET:

Code Details

200

Response body

```
{
  "id": 1,
  "customerId": "C001-001",
  "subject": "Password reset link not working",
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth",
  "status": "COMPLETED",
  "failureReason": null,
  "analysis": {
    "category": "ACCOUNT_ACCESS",
    "summary": "Customer reported an issue with subject \"Password reset link not working\" related to Auth.",
    "suggestedResponse": "Thank you for reaching out. We've received your report and our account security team is looking into it. We'll follow up as soon as we have an update.",
    "recommendedTeam": "ACCOUNT_SECURITY",
    "confidence": 0.85
  },
  "createdAt": "2026-09-27T11:19:08.435Z",
  "updatedAt": "2026-09-27T11:19:08.829Z"
}
```

Response headers

```
connection: keep-alive
content-type: application/json
date: Sat, 27 Sep 2025 11:20:08 GMT
keep-alive: timeout=60
transfer-encoding: chunked
```

Responses

GET /api/tickets/{id}

Parameters

Name Description

id `required`

Integer (int32)

(path)

1

Execute Clear

Responses

Code Details

200

Response body

PERSISTENCE VERIFICATION — H2 CONSOLE

Querying `SELECT * FROM TICKETS` against the in-memory H2 database confirms both the ticket data and the AI-generated analysis are persisted correctly end to end:

jdbc:h2:mem:ticketdb

TICKETS

INFORMATION_SCHEMA

Users

H2 2.2.224 (2023-09-17)

Run Run: Selected Auto complete Clear SQL statement:

SELECT * FROM TICKETS

CONFIDENCE	ID	CREATED_AT	UPDATED_AT	SUMMARY	FAILURE_REASON	SUGGESTED_RESPONSE
0.85	1	2026-09-27 11:19:08.435242+00	2026-09-27 11:19:08.829127+00	Customer reported an issue with subject "Password reset link not working" related to Auth.	null	Thank you for reaching out. We've received
0.9	2	2026-09-27 12:40:06.879457+00	2026-09-27 12:40:06.987279+00	Customer reported an issue with subject "Unable to login" related to Auth.	null	Thank you for reaching out. We've received

(2 rows, 13 ms)

Edit

DEVELOPMENT NOTES

Approximate time spent on the assessment: 7–8 hours.

AI tools used during development: roughly 25% of the development process involved AI assistance, used as a supporting productivity tool for:

- Clarifying error messages and debugging issues during development.
- Reviewing and improving test cases, and identifying additional edge cases.
- Generating sample request payloads and test data for API testing.
- Checking syntax and suggesting minor code-level improvements.
- Troubleshooting configuration and environment-related issues.

ASSUMPTIONS & LIMITATIONS

- Each request creates exactly one ticket — there is no bulk or batch ticket submission.
- One LLM call produces one complete analysis; there is no support for partial, streaming, or multi-turn analysis.
- Persistence: H2 runs in-memory (`ddl-auto: create-drop`), so all ticket data is lost on every restart. This was a deliberate choice so the project runs with zero external setup, not an oversight.
- No authentication or authorization — both endpoints are fully open. Anyone with network access to the service can create or read any ticket.
- No retry or backoff on transient LLM failures — a timeout or transient provider error currently fails the ticket immediately rather than retrying with backoff.
- Next step: add API versioning and a stable, documented contract (once OpenAPI generation is fixed) before external consumers depend on this service.